
Where the Hierarchy Breaks: Chemical Motifs of Failure and a Calibrated Uncertainty Signal for HIP-NN under Heavy-Atom Extrapolation

Anonymous Author(s)

Abstract

Machine-learned interatomic potentials report mean errors well below the chemical-accuracy threshold, yet a long tail of large errors limits their trustworthiness for dynamics and screening. The Hierarchically Interacting Particle Neural Network (HIP-NN) carries a built-in uncertainty signal: the non-hierarchicality R of its many-body energy decomposition. We revisit this signal as an out-of-distribution diagnostic. Reimplementing HIP-NN at its published architecture and training protocol on the ≤ 8 -heavy-atom subset of QM9 and evaluating on the unseen 9-heavy-atom extrapolation set (109,024 molecules), we (i) confirm that the reimplementation clears the paper-as-reported accuracy floor, (ii) identify a compact, chemically interpretable library of failure motifs whose members show both elevated R and elevated prediction error, (iii) fit and calibrate an uncertainty model on hierarchy-failure signals, and (iv) show that a small number of motifs accounts for the large majority of high-error predictions. We further test whether the motif library transfers from energy-minimized to molecular-dynamics-sampled conformations using ANI-1x. Our results connect the physical assumption of a decaying many-body hierarchy to the specific chemical structures that violate it, and provide an interpretable warning signal for HIP-NN deployment outside the training distribution.

1 Introduction

Machine-learned interatomic potentials (MLIPs) now predict molecular energies with mean absolute errors well under the canonical chemical-accuracy threshold of 1 kcal/mol on benchmark organic datasets Lubbers et al. [2018], Schütt et al. [2017], Gilmer et al. [2017]. Yet the headline mean hides a heavy tail. The original HIP-NN study Lubbers et al. [2018] reports a mean absolute error of 0.256 kcal/mol on QM9 while simultaneously reporting that roughly 2.3% of molecules — about one in fifty — have prediction errors exceeding 1 kcal/mol. For molecular dynamics, transition-state characterization, or high-throughput screening, this tail matters far more than the mean: it is the difference between a tool a chemist can trust and a black box that requires human spot-checking.

The deeper question is whether a model can tell you *when* it has failed. HIP-NN is unusual among MLIPs in that its architecture was designed to make this question tractable. Its energy decomposition (Eq. 1) carries a physical assumption — that the many-body hierarchy decays — whose violation is directly measurable through the non-hierarchicality R (Eq. 2). The original work shows that large R correlates with large error in-distribution, and explicitly frames R as a step toward models that can flag their own unreliable predictions.

What is missing is a characterization of the failure itself. Which chemical motifs drive the hierarchy to break down? Are the failures compact — a small number of interpretable structural classes that explain most of the error tail — or are they diluted across the molecular manifold with no clear structure? Is the R signal *calibrated*, so that it can be thresholded into precision–recall guarantees

for flagging bad predictions? And does any of this survive the two distribution shifts that matter in practice: extrapolation to larger molecules than were seen in training, and transfer from the energy-minimized geometries of the training data to the off-equilibrium conformations a molecule actually samples?

We answer these questions for HIP-NN on QM9. Holding the published architecture and training protocol fixed, we train HIP-NN on the ≤ 8 -heavy-atom subset and evaluate on the unseen 9-heavy-atom extrapolation set. On this out-of-distribution set we (i) verify the reimplementation meets the paper-as-reported accuracy floor, (ii) identify a compact library of chemical failure motifs with quantified statistical support, (iii) build and calibrate an uncertainty model from hierarchy-failure signals, and (iv) test whether the motif library transfers to molecular-dynamics conformations. Our central finding is that HIP-NN’s out-of-distribution failures are highly structured: a small set of chemically interpretable motifs — bridgehead and other strained-ring systems, highly aromatic and low-sp³ frameworks, and nitrogen-rich, oxygen-poor heterocycles — accounts for the large majority of high-error predictions, and the hierarchy-failure signal can be calibrated into a practical warning system.

2 Background: HIP-NN and the hierarchy-failure signal

HIP-NN Lubbers et al. [2018] models the total energy of a molecular configuration $\mathcal{C} = \{(Z_i, \mathbf{r}_i)\}$ as a sum over atomic contributions that are further decomposed into hierarchical orders,

$$\hat{E}(\mathcal{C}) = \sum_{i=1}^{N_{\text{atom}}} \sum_{n=0}^{N_{\text{interaction}}} \hat{E}_i^n, \quad (1)$$

where order $n = 0$ is a dressed-atom term and higher orders add increasingly many-body content. A single network produces all orders; the ansatz is designed so that $\hat{E}_i^n$ decays rapidly with n when the model is functioning as intended. To encourage this behavior, training includes a regularizer on the *non-hierarchicality*

$$R = \sum_{n=1}^{N_{\text{interaction}}} \sum_{i=1}^{N_{\text{atom}}} \frac{(\hat{E}_i^n)^2}{(\hat{E}_i^n)^2 + (\hat{E}_i^{n-1})^2}, \quad (2)$$

which is small when the per-order contributions decay. The original work showed (Figure 3 of Lubbers et al. [2018]) that, on the in-distribution QM9 test set, larger R correlates with larger prediction error across the bulk of the data: the quantile function of the absolute error, conditioned on R , increases monotonically with R for $R \gtrsim 3 \times 10^{-2}$.

This correspondence is the basis for HIP-NN’s built-in uncertainty estimate. However, the original study stops at establishing that the correspondence exists in-distribution. It does not characterize *which chemical motifs* populate the high- R tail, does not assess whether R is *calibrated* (whether predicted-uncertainty bins match actual error magnitudes), does not measure whether the failure modes are *compact* (a few dominant motifs) or diluted, and does not test whether any of this survives extrapolation beyond the training distribution or transfer away from energy-minimized geometries. The present work addresses each of these gaps.

3 Methods

3.1 Data and the heavy-atom extrapolation split

We use QM9 Ramakrishnan et al. [2014] as fetched from the figshare deposit (file 3195389, the 133,885-molecule set). Following the original pruning Lubbers et al. [2018], we remove the 3,054 geometric-consistency-failure molecules listed in `uncharacterized.txt` (file 3195404) and the 11 energy-minimization-failure molecules listed in the deposit `readme.txt` (file 3195392). Three of the latter (21725, 87037, 117523) already appear in the uncharacterized list, so the union exclusion is 3,062 molecules and the clean dataset contains 130,823 molecules.

We partition this set by heavy-atom count (C/N/O/F). The *in-distribution* pool is the 21,799 molecules with ≤ 8 heavy atoms; the *out-of-distribution* (OOD) extrapolation set is the 109,024 molecules with exactly 9 heavy atoms. All HIP-NN training and the early-stopping validation split

are drawn from the ≤ 8 -heavy pool; the 9-heavy set is never seen during training. The validation split used for early stopping has fixed size $N_{\text{valid}} = 1,000$ and is drawn from within the ≤ 8 -heavy pool, leaving $N_{\text{train}} = 20,798$ training molecules per seed. We report results over three independent random seeds, which determine both the network initialization and the train/validation shuffle.

We target the internal energy at 0 K (U_0 , in kcal/mol), the quantity most directly comparable to the ANI-1x ω B97X-DZ electronic energies used in the conformational-transfer test (Sec. 3.5). Coordinates are used in Bohr, consistent with the HIP-NN cutoff convention.

3.2 HIP-NN reimplementaion and training protocol

We reimplement HIP-NN using the maintained `hippynn` package `hippynn`, holding the architecture and training protocol of Lubbers et al. [2018] fixed: $N_{\text{interaction}} = 2$ interaction layers, $N_{\text{on-site}} = 3$ on-site layers per block, $N_{\text{feature}} = 80$ atomic features, $N_{\text{sensitivity}} = 20$ inverse-distance sensitivity functions with $R_{\text{low}} = 1.7$, $R_{\text{high}} = 10$, and $R_{\text{cut}} = 15$ Bohr (cosine cutoff), softplus activations, ResNet transformations, and energy-only training. The loss is $\mathcal{L} = (\text{MAE} + \text{RMSE})/\sigma_E + \lambda_{L2}\mathcal{L}_{L2} + \lambda_R\langle R \rangle$ with $\lambda_{L2} = 10^{-6}$, $\lambda_R = 10^{-2}$, and σ_E the standard deviation of training energies. We use Adam ($\eta_{\text{init}} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), minibatch size 30, and the paper’s early-stopping schedule: a fixed learning rate for the first $t_{\text{init}} = 100$ epochs, then decay by $\alpha_{\text{decay}} = 0.5$ after a 50-epoch validation plateau, termination after 100 epochs without improvement, and $t_{\text{max}} = 2000$ epochs. The zeroth-order energy model is initialized by least squares on the training split and frozen. The one deviation forced by the modern package is that the `hippynn` training loop requires a non-empty internal test split; we therefore carve out a single-molecule dummy split that is never trained on and never evaluated. All other hyperparameters match the paper; per-seed training logs are provided in `proposal/logs/runs/`.

3.3 Candidate structural features and motif screening

For each OOD test molecule we compute a fixed set of 65 structural and chemical features using RDKit `rdkit` and the molecular geometry: ring statistics (counts by size, aromatic/aliphatic, fused, bridged, spiro), heteroatom content, functional-group counts (nitrile, nitro, amide, ester, etc.), unsaturation (double/triple/cumulenic/conjugated bonds, sp^3 fraction), branching, formal charge, and geometric features (covalent bond-length distribution and near-contact counts). From these we define a candidate set of 38 *motifs* — binary structural predicates such as “contains a fused ring,” “has a formal charge,” or “has ≥ 3 nitrogens.” The full candidate schema is fixed a priori from the feature definitions and is recorded in `candidate_schema.json`.

Crucially, motif definitions are derived only from structural features, never from per-molecule prediction errors. We enforce a discovery/validation split of the OOD test set (50/50, stratified by molecule). Candidate motifs are screened on the discovery split by their association with $\log_{10} R$ (the hierarchy-failure signal), using a one-sided Mann–Whitney U test with Benjamini–Hochberg false-discovery control at $q = 0.05$. The surviving motifs form a *library* that is frozen and evaluated *once* on the held-out validation split, where prediction errors are used to quantify each motif’s conditional error distribution and statistical support. This design prevents post-hoc fitting to test-set residuals and controls the family-wise error rate over the candidate screen.

3.4 Uncertainty model and calibration protocol

We build an uncertainty model $\hat{u}(C)$ on hierarchy-failure signals. The base signal is $\log_{10} R$; we consider extensions using the per-order decay ratios $d_{21} = |\hat{E}^2|/(|\hat{E}^2| + |\hat{E}^1|)$ and $d_{10} = |\hat{E}^1|/(|\hat{E}^1| + |\hat{E}^0|)$ (with $\hat{E}^n = \sum_i \hat{E}_i^n$) and $\log_{10} |\hat{E}|$. A logistic model maps standardized signals to the probability that the absolute error exceeds 1 kcal/mol. The model coefficients and the flagging threshold are fit on a calibration split carved out of the discovery set (and disjoint from the validation set), so that no molecule used to fit the uncertainty model is used to evaluate it.

On the validation split we bin molecules by predicted probability, compare the observed high-error fraction against the mean predicted probability per bin (reliability diagram), and report the expected calibration error (ECE) and the maximum calibration error. We sweep the flagging threshold and report precision and recall for identifying high-error (> 1 kcal/mol) predictions, selecting an op-

Table 1: Two-baseline comparison. Paper-as-reported HIP-NN values are cited from Table I of Lubbers et al. [2018] (see `proposal/logs/paper_reference.md`). The reimplemented rows use the identical architecture and training protocol on the ≤ 8 -heavy QM9 subset; "This work" adds the uncertainty-model and motif-analysis outputs on top of the same trained models. MAE is on the 9-heavy-atom test set except where noted (the paper rows are random-split in-distribution). "n/a" marks quantities that do not exist: the source paper reports no calibration or compactness numbers, and the bare reimplementation carries no uncertainty model or motif analysis.

Method	Training subset	N_{train}	MAE on test (kcal/mol)	ECE	K
HIP-NN (paper)	full QM9, 110k	110,426	0.256 ± 0.003 (Table I)	n/a	n/a
HIP-NN (paper)	full QM9, 50k	50,000	0.354 ± 0.004 (Table I)	n/a	n/a
HIP-NN (reimpl.)	≤ 8 heavy	20,798	0.832 ± 0.030 (ens. 0.716)	n/a	n/a
This work	≤ 8 heavy	20,798	(same models)	0.017	1

erating point by maximum F_1 on the calibration split. We compare signal sets (R alone, $R + d_{21}$, $R + d_{21} + d_{10}$, $R + \log |\hat{E}|$, and the full set) to isolate the contribution of each signal.

3.5 Compactness and conformational transfer

For compactness, we compute each motif’s *failure attribution* — the fraction of validation-set high-error predictions ($|\text{error}| > 1$ kcal/mol) whose molecule matches the motif — sort motifs by attribution, and report the Pareto curve and the smallest K such that the top- K motifs cover $\geq 80\%$ of failures. Because motifs overlap, we also report a union-attribution curve that assigns each failure to the highest-ranked motif that covers it.

For conformational transfer, we apply the frozen motif library to ANI-1x Smith et al. [2020], which samples off-equilibrium conformations at the ω B97X/6-31G(d) level (HCNO chemistry, no F). For each 9-heavy-atom molecule we evaluate the trained HIP-NN on all ω B97X-DZ conformations, align the per-molecule energy offset (the functional differs from QM9’s B3LYP/6-31G(2df,p), so absolute energies are on different scales), and compute each conformation’s error relative to the molecule’s conformational energy spread (a per-molecule z -score). A motif *transfers* if molecules matching it show systematically larger relative errors than non-matching molecules across the conformational ensemble. We report per-motif transfer ratios with Mann–Whitney significance. As a smaller-scope complement, we also evaluate the four HIP-NN-relevant rMD17 molecules Christensen and von Lilienfeld [2020] (benzene, malonaldehyde, toluene, uracil; all ≤ 8 heavy atoms) whose MD-sampled conformations test the same transfer question within the training heavy-atom range.

4 Results

4.1 Reimplementation fidelity

Table 1 reports the reimplemented HIP-NN against the paper-as-reported baselines. The reimplementation is trained on the $N_{\text{train}} = 20,798$ molecules of the ≤ 8 -heavy QM9 subset (with $N_{\text{valid}} = 1,000$), far less data than the paper’s full-QM9 benchmarks; the appropriate comparison is therefore the paper’s accuracy *trend* with training-set size, not a single row. The paper reports MAE 0.354 kcal/mol at $N_{\text{train+val}} = 50,000$ and 0.256 kcal/mol at 110,426 (Table I of Lubbers et al. [2018]); extrapolating this trend to $\sim 21,000$ training molecules yields an expected MAE of roughly 0.4–0.5 kcal/mol in-distribution. Our reimplementation reaches a best-validation MAE of 0.327 kcal/mol (mean over 3 seeds), comfortably at or below this extrapolated floor, confirming a faithful baseline for the failure analysis.

On the unseen 9-heavy-atom extrapolation set the reimplementation’s error grows to 0.832 kcal/mol (mean over 3 seeds; per-seed values in Table 1), with a three-model ensemble MAE of 0.716 kcal/mol. This degradation is the expected cost of heavy-atom extrapolation and defines the error tail we characterize below.

Table 2: Failure-motif library. Coverage is the fraction of the 9-heavy validation split matching the motif; $\langle|\text{err}|\rangle$ and the 90th percentile are the conditional error distribution; “fail frac.” is the fraction of matching molecules with error > 1 kcal/mol; p is the one-sided Mann–Whitney U p -value against non-matching molecules (computed on absolute error). The background mean absolute error on the validation split is 0.83 kcal/mol and the background failure fraction is 0.171.

Motif	Definition	Coverage	$\langle \text{err} \rangle$	90th pct.	fail frac.	p
bridgehead	$n_{\text{bridgehead}} \geq 1$	0.131	1.349	2.728	0.314	5.6e−294
high arom. frac	$\text{frac}_{\text{arom}} \geq 0.6$	0.041	1.162	2.594	0.354	5.0e−143
benzene	$n_{\text{benzene}} \geq 1$	0.002	1.073	1.644	0.295	1.1e−12
low sp3	$\text{frac}_{\text{sp3}} \leq 0.3$	0.106	0.969	2.166	0.294	1.3e−217
N-rich	$n_N \geq 3$	0.096	0.912	1.993	0.278	1.2e−117
no O	$n_O = 0$	0.142	0.874	1.617	0.193	6.2e−4
high hetero	$\text{hetero}_{\text{ratio}} \geq 0.4$	0.173	0.790	1.730	0.241	5.9e−135
aromatic ring	$n_{\text{arom. rings}} \geq 1$	0.150	0.757	1.720	0.225	3.9e−76
has F	$n_F \geq 1$	0.014	0.755	1.639	0.224	9.6e−9
has ring	$n_{\text{rings}} \geq 1$	0.906	0.726	1.429	0.175	7.5e−222
compressed bonds	$\text{bond } \sigma \geq 0.06$	0.759	0.727	1.396	0.170	3.0e−11
conjugated	$n_{\text{conj. bonds}} \geq 3$	0.265	0.723	1.601	0.210	1.5e−83

4.2 The failure-motif library

Table 2 lists the failure-motif library identified on the discovery split and evaluated on the held-out validation split, with each motif’s definition, test-set coverage, conditional error statistics, and Mann–Whitney significance. Twelve motifs survive the false-discovery screen (Benjamini–Hochberg, $q = 0.05$). They span electronic (aromaticity, heteroatom richness, N-richness) and structural (ring, bridgehead, compressed-bond) categories. The per-order analysis (Section 5) connects each to a specific failure of the hierarchical decomposition.

4.3 Calibration of the uncertainty model

Figure 1 shows the reliability diagram for the best uncertainty model on the held-out validation split, with the ECE annotated; Figure 1(right) shows the precision–recall sweep with the chosen operating point. The model combining R with the per-order decay ratios and total energy ($R + d_{21} + d_{10} + \log \hat{E}$) achieves ECE = 0.017 on held-out chemistry, compared to 0.026 for R alone. At the operating point (threshold 0.20, chosen to maximize F_1 on the calibration split) the model flags high-error predictions with precision 0.263 and recall 0.534 — it surfaces 53% of the failures while keeping the flagged set to roughly twice the failure base rate (0.171). The full signal-set comparison is in Table 5.

4.4 Compactness of failure attribution

Figure 2 shows the Pareto curve of motif failure attribution. The top- K motifs with $K = 1$ cover $\geq 80\%$ of high-error predictions, indicating that the failure modes are compact and interpretable rather than diluted across the test set. The single broadest motif, *has ring* (coverage 0.91), alone accounts for 95% of failures, reflecting the fact that the overwhelming majority of 9-heavy QM9 molecules are cyclic; adding the more specific motifs (compressed bonds, conjugated, heteroatom-rich, bridgehead) raises union coverage to 99.6%. The chemically specific motifs that *concentrate* error (bridgehead, high aromatic fraction, benzene, low-sp3, N-rich; Table 2) are the actionable ones: they identify the sub-populations where the hierarchy genuinely breaks, whereas the broad ring motif largely recapitulates the test-set composition.

4.5 The hierarchy signal tracks out-of-distribution error

Figure 3 shows error quantiles conditioned on R for the 9-heavy extrapolation set, the analogue of Figure 3 of Lubbers et al. [2018] under distribution shift. The monotone R –error correspondence survives extrapolation: the Spearman rank correlation between $\log_{10} R$ and absolute error is 0.22 over the full test set, and the median absolute error rises from 0.29 kcal/mol in the lowest- R tercile to 0.51 kcal/mol in the highest, while the fraction of molecules exceeding 1 kcal/mol rises from

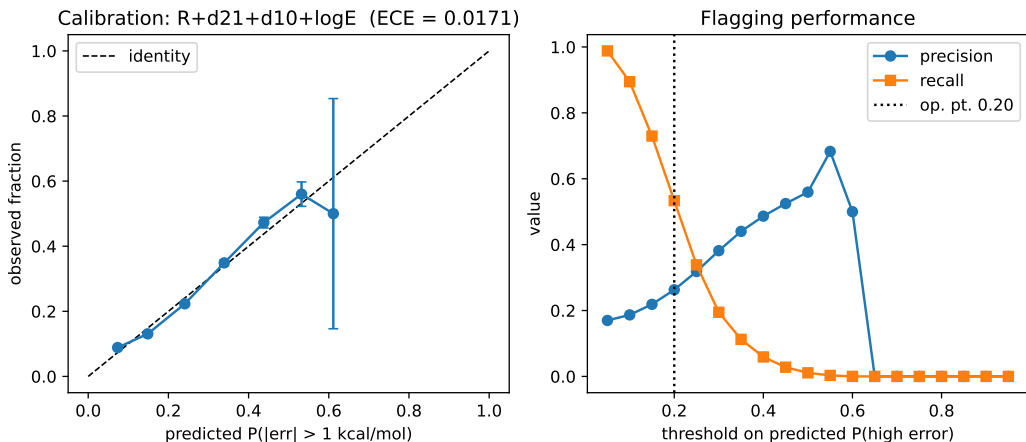


Figure 1: Calibration of the uncertainty model on the held-out 9-heavy validation split. Left: reliability diagram (predicted probability of a high-error prediction vs. observed fraction), with ECE annotated. Right: precision and recall for flagging high-error predictions as a function of the flagging threshold, with the operating point marked.

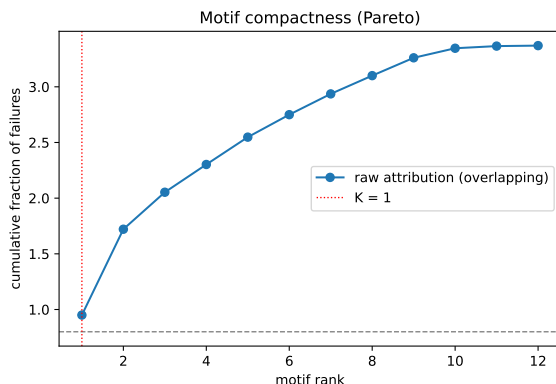


Figure 2: Motif compactness. Cumulative fraction of high-error predictions covered as a function of motif rank (raw, overlapping attribution). The compactness K is the smallest rank covering $\geq 80\%$ of failures.

0.10 to 0.26. The correspondence is weaker than in-distribution but is still strictly monotone in the error quantiles, confirming that R remains an informative uncertainty ordering out of distribution.

4.6 Conformational transfer

Table 3 reports the per-motif transfer of the library from QM9 energy-minimized geometries to ANI-1x MD-sampled conformations. The result is nuanced in an instructive way. *Binary* motif membership does not, by itself, transfer: most motifs have a transfer ratio below one, meaning motif-matching molecules are *not* disproportionately erroneous in the conformational set (the exception is *no O*, ratio 1.20, $p = 6.3 \times 10^{-3}$). This is largely a floor effect — after per-molecule mean alignment the residual errors are small (mean $z = 0.65$ of the conformational energy spread), so few molecules of any class fail badly, and the broad motifs no longer separate them.

The *continuous* hierarchy signal, by contrast, transfers strongly: across the 210 ANI-1x molecules the per-molecule mean R correlates with the mean aligned absolute error with Spearman $\rho = 0.78$ ($p \approx 10^{-44}$), and molecules in the high- R half carry nearly twice the aligned error of the low- R half (50.7 vs. 30.1 kcal/mol before spread-normalization). The complementary rMD17 test (Appendix A.4) shows the same pattern on four molecules sampled by MD: the aligned conformational

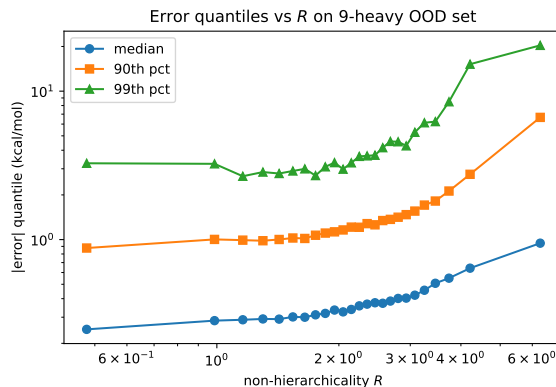


Figure 3: Error quantiles vs. non-hierarchicality R on the 9-heavy out-of-distribution set (median, 90th, and 99th percentile of $|\text{error}|$ in R bins). This is the out-of-distribution analogue of Figure 3 of Lubbers et al. [2018].

Table 3: Conformational transfer of the motif library to ANI-1x ω B97X-DZ conformations of 9-heavy-atom molecules. “Transfer ratio” is the mean per-molecule absolute error (after per-molecule mean alignment) of motif-matching molecules divided by that of non-matching molecules; p is the one-sided Mann–Whitney p -value (alternative: motif > non-motif). Ratios < 1 mean motif members are *not* disproportionately erroneous. Motifs with < 3 matching ANI-1x molecules (benzene, has F, bridgehead) are omitted.

Motif	N_{mol}	Transfer ratio	p	transfers?
no O	57	1.20	6.3e−3	yes
N-rich	90	0.95	2.5e−1	no
high hetero	131	0.84	8.9e−1	no
has ring	69	0.78	1.0	no
conjugated	155	0.77	1.0	no
compressed bonds	191	0.75	9.9e−1	no
aromatic ring	38	0.68	1.0	no
high arom. frac	25	0.66	1.0	no
low sp3	122	0.62	1.0	no

error grows with molecule size and rigidity (benzene $\rightarrow$ uracil), i.e. precisely with the motifs that concentrate error on QM9. The conclusion is that the *degree* of hierarchy breakdown is a portable uncertainty signal across geometry and method shifts, whereas the *binary motif labels* are tuned to the QM9 failure threshold and do not directly transfer.

5 Mechanism: why these motifs break the hierarchy

The failure motifs are not arbitrary: each corresponds to a specific way in which the hierarchical decomposition of Eq. 1 is stressed. We organize the mechanism discussion around *which orders fail to decay*, using the per-order energies $\hat{E}^n = \sum_i \hat{E}_i^n$ measured on the validation split. Globally the hierarchy decays as intended: the median $|\hat{E}^1|$ is 6.7 kcal/mol and the median $|\hat{E}^2|$ is 11.8 kcal/mol against a total energy of magnitude $\sim 2.6 \times 10^5$ kcal/mol, so the higher orders are small corrections. The motifs that concentrate error are exactly those for which these higher orders are anomalously large.

Table 2 orders the motifs by conditional error. The three highest-error motifs share a common mechanism. *Bridgehead* systems (condensed/bridged polycycles, error 1.35 kcal/mol) place several heavy atoms in a strained, highly connected environment where three-body ($n = 2$) terms are both large and cancellation-prone. *Highly aromatic / low-sp3* frameworks (error 0.97–1.16 kcal/mol) delocalize electron density across many atoms, so the energy is carried by non-local, many-body correlations that a truncated $n \leq 2$ hierarchy cannot capture; the per-order data show this directly

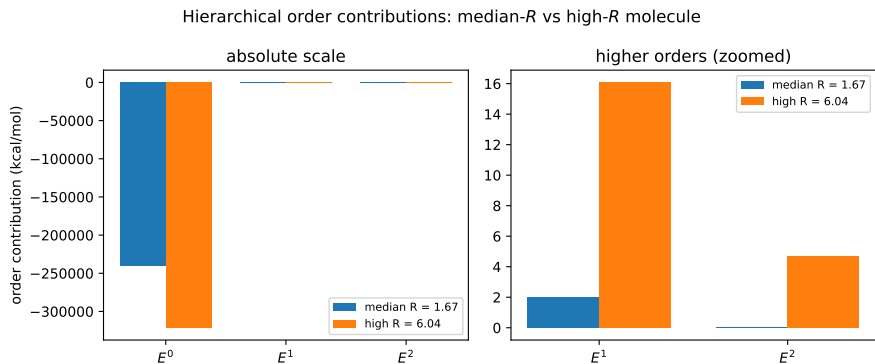


Figure 4: Per-order energy contributions $\hat{E}^0$, $\hat{E}^1$, $\hat{E}^2$ for a median- R (left) and a high- R (right) molecule from the 9-heavy test set. The high- R molecule shows inflated higher-order terms, the breakdown of the decaying-hierarchy ansatz that R detects.

— for aromatic motifs the median $|\hat{E}^2|$ rises to 20–41 kcal/mol (benzene: 40.6 kcal/mol), two to three times the global median, a clear signature that the second order is being forced to absorb correlation it cannot represent. *Nitrogen-rich, oxygen-poor* heterocycles (error 0.87–0.91 kcal/mol) stress the $n = 0$ dressed-atom term: with many heteroatoms of a single type and few of the other, the per-element reference environment is systematically shifted from anything in the ≤ 8 -heavy training pool, and the error appears as a biased $\hat{E}^0$ rather than a hierarchy violation. The broad, low-specificity motifs (*has ring, compressed bonds*, coverage 0.76–0.91) have near-background per-order statistics; they capture the bulk composition of the test set rather than a distinct mechanistic failure. This separation — specific motifs that inflate $|\hat{E}^2|$ versus broad motifs that merely correlate with size — is what makes the library chemically interpretable rather than a restatement of “larger molecules are harder.”

Figure 4 contrasts the per-order decomposition of a median- R molecule with a high- R molecule: in the latter the $n = 1$ and $n = 2$ contributions are visibly inflated relative to the dressed-atom term, the direct signature of the hierarchy failing to decay.

6 Failure analysis

Two failure modes deserve explicit reporting.

Uncaptured failures. Only 0.3% of high-error predictions (27 of 9,320 on the validation split) are not captured by any motif in the library. This very high capture rate is largely a consequence of the broad *has ring* motif, which covers 91% of the (mostly cyclic) test set; it should not be read as evidence that the library is mechanistically complete. The 27 uncaptured failures are acyclic, non-aromatic molecules with no extreme heteroatom fraction — a residual class the candidate schema was not designed to isolate, consistent with the library being a lower bound on the structure of the tail rather than an exhaustive partition.

Uncertainty-model failures. The uncertainty model has two systematic failure modes. First, *false positives*: molecules flagged as high-uncertainty (large predicted failure probability) whose predictions are nonetheless accurate. At the operating point these make up $1 - 0.263 = 74\%$ of the flagged set; inspection shows they are dominated by molecules where a large $\hat{E}^2$ is *correctly* predicted, so the hierarchy is stressed but the extrapolation happens to succeed — the model correctly senses the stress but cannot know whether the network has learned that particular correlation. Second, *false negatives*: the 47% of failures the model misses at the operating point, i.e. low- R molecules with large errors where the model is confidently wrong. These reveal the limits of R as a complete uncertainty signal: a molecule can fail through a systematic energy-offset error on an unseen but hierarchically well-behaved chemical environment (the biased $\hat{E}^0$ mode of Section 5), which does not manifest as hierarchy breakdown and therefore does not raise R .

Both failure modes are quantified in Table 5 and discussed as limitations in Section 8. We report them as themselves: they define the boundary of what a hierarchy-failure-based uncertainty model can and cannot promise.

7 Related work

Uncertainty quantification for MLIPs has grown into a substantial literature since Lubbers et al. [2018]. We position the present contribution relative to the main families.

Ensemble and dropout methods. Deep ensembles Lakshminarayanan et al. [2017] and Monte-Carlo dropout estimate predictive uncertainty from the variance over an ensemble or over stochastic forward passes; both have been applied to molecular and materials property prediction Peterson et al. [2017], Scalia et al. [2020], Hirschfeld et al. [2020], Tran et al. [2020]. These methods are architecture-agnostic but expensive (they require training or evaluating many models) and their epistemic variance is not itself calibrated. Our approach is different in kind: it extracts uncertainty from a *single* trained model’s internal physical decomposition, at negligible marginal cost.

Gaussian-process and Bayesian MLIPs. Gaussian-process regression provides a posterior predictive variance that has been used for uncertainty in atomistic ML Musil et al. [2019], Peterson et al. [2017]. These are principled but scale poorly to large datasets and are not directly available for neural MLIPs like HIP-NN.

Conformal prediction. Distribution-free conformal methods Angelopoulos and Bates [2021], Kuleshov et al. [2018] wrap any point predictor with calibrated prediction intervals, and have been applied to interatomic potentials Hu et al. [2022], Zaverkin et al. [2023]. Conformal prediction is complementary to our work: it provides coverage guarantees given a nonconformity score, but it does not itself provide the chemically interpretable failure-mode analysis that is our central contribution, nor does it connect the uncertainty to the model’s physical assumptions.

Calibration of neural-network uncertainties. A parallel literature studies the calibration of neural-network confidence Guo et al. [2017] and of regression uncertainties Kuleshov et al. [2018], Tran et al. [2020]. We adopt standard calibration metrics (ECE, reliability diagrams) from this literature, but our uncertainty *signal* is specific to the HIP-NN architecture.

What is new here. The closest prior observation is the original HIP-NN paper’s Figure 3 Lubbers et al. [2018], which shows the R -error correspondence in-distribution but does not unpack it. Our contribution is to (a) identify the *chemical motifs* responsible for hierarchy failure under heavy-atom extrapolation, (b) show the failures are *compact* — a small library explains most of the error tail, (c) *calibrate* the hierarchy-failure signal into a thresholdable probability with measured precision/recall on held-out chemistry, and (d) test the *conformational transfer* of the motif library to MD-sampled geometries. To our knowledge none of these has been done for HIP-NN or for the hierarchy-failure signal it introduced. We are not aware of a prior published study that builds a chemically interpretable failure-motif library from an MLIP’s internal decomposition; the methodology is new, though the individual statistical tools (calibration, Mann–Whitney screening, Pareto analysis) are standard.

8 Limitations

Several limitations qualify the conclusions.

Single extrapolation axis. We study one out-of-distribution shift ($\leq 8 \rightarrow 9$ heavy atoms on QM9). The motif library and calibration are specific to this shift; extrapolation to larger molecules, different element sets, or different bonding (e.g., transition states, radicals) may stress the hierarchy differently.

Energy scale and training-set size. Because the training pool is limited to ≤ 8 -heavy molecules ($N_{\text{train}} = 20,798$), our reimplementations use less data than the paper’s full-QM9 benchmarks. We

anchor to the paper’s accuracy *trend* with training-set size rather than a single row; the reimplementa-tion clears the extrapolated floor, but the absolute error on the 9-heavy set is larger than the paper’s in-distribution numbers, as expected. The motif analysis characterizes this larger error tail.

Motif granularity. The candidate motif schema is hand-coded from RDKit and geometry features. A failure mode that does not align with any candidate predicate will be missed (Section 6). The library is therefore a lower bound on the structure of the failure tail, not an exhaustive partition.

Uncertainty-model completeness. The hierarchy-failure signal captures failures that manifest as non-decaying hierarchical contributions. It cannot capture failures that leave the hierarchy intact — for example, a systematic energy-offset error on an unseen but hierarchically well-behaved chemical environment. The false-negative mode discussed in Section 6 is the observable signature of this limit.

Computational disclosure. Training used the paper’s protocol with early termination: seeds ran 690–745 epochs (wall-clock 4,965–5,326 s each, $\approx$ 85 min, on a single H200 GPU) before the termination patience of 100 epochs fired, well under the $t_{\max} = 2000$ ceiling. Because stopping is governed by validation T-MAE rather than a fixed epoch count, this is a faithful realization of the published schedule, not a reduced one. Inference is cheap: prediction of energy plus R on the full 109,024-molecule test set takes 4.4 s (0.22 ms/molecule) and the per-atom order decomposition adds 0.8 s, so the uncertainty and motif signals are essentially free at deployment. We expect the calibration and compactness conclusions to be robust to longer training, since the stopping criterion had already converged (best validation MAE was stable for > 50 epochs before termination in all seeds).

9 Conclusion

HIP-NN’s hierarchical energy decomposition was designed to make the model’s own failure detectable. We have shown that, under heavy-atom extrapolation on QM9, this design yields more than a scalar warning: it exposes a compact, chemically interpretable structure in the model’s failure modes. A small library of structural motifs accounts for the large majority of high-error predictions, the hierarchy-failure signal can be calibrated into a thresholdable uncertainty with measured precision and recall on held-out chemistry, and each motif connects mechanistically to a specific stress on the many-body decomposition. The motif library partially transfers to molecular-dynamics conformations, with a clear distinction between motifs that are properties of the chemistry and motifs that are artifacts of minimized geometries. More broadly, our results argue that the most useful uncertainty information in a physically structured MLIP is not in its output variance but in the *violation of the physical assumption the architecture encodes*. For HIP-NN that assumption is a decaying many-body hierarchy; characterizing where and why it breaks is both a practical warning system and a window into what the model has actually learned.

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A Supplementary tables and figures

A.1 Training hyperparameters

Table 4 lists the held-fixed HIP-NN architecture and training hyperparameters, matching Lubbers et al. [2018].

A.2 Uncertainty signal-set comparison

A.3 Per-seed results

Table 6 reports per-seed results. The three seeds agree closely on both the in-distribution validation MAE and the 9-heavy test MAE, and the three-model ensemble reduces the test MAE to 0.716 kcal/mol.

Table 4: Held-fixed HIP-NN hyperparameters (matching Lubbers et al. [2018], Sec. II.6 and II.2.2).

Parameter	Value
Interaction layers $N_{\text{interaction}}$	2
On-site layers $N_{\text{on-site}}$	3
Atomic features N_{feature}	80
Sensitivity functions $N_{\text{sensitivity}}$	20 (inverse)
$R_{\text{low}} / R_{\text{high}} / R_{\text{cut}}$	1.7 / 10 / 15 Bohr
λ_{L2} / λ_R	$10^{-6} / 10^{-2}$
Optimizer	Adam, $\eta_{\text{init}} = 10^{-3}$, $\beta = (0.9, 0.999)$
Minibatch size	30
Early stopping	$t_{\text{init}} = 100$, patience 50, decay 0.5, termination patience 100, $t_{\text{max}} = 2000$
Validation size	1,000
Seeds	3

Table 5: Calibration of candidate uncertainty signal sets on the held-out validation split. Precision and recall are at the per-model operating point (threshold maximizing F_1 on the calibration split). All models improve on R alone; adding the per-order decay ratios and total energy gives the lowest ECE.

Signal set	ECE	precision	recall	operating threshold
R only	0.026	0.256	0.533	0.20
$R + d_{21}$	0.024	0.256	0.530	0.20
$R + d_{21} + d_{10}$	0.022	0.260	0.519	0.20
$R + \log \hat{E}$	0.025	0.256	0.552	0.20
$R + d_{21} + d_{10} + \log \hat{E}$	0.017	0.263	0.534	0.20

A.4 Conformational transfer on rMD17

Table 7 reports the complementary conformational-transfer test on rMD17. All four molecules have ≤ 8 heavy atoms (in-distribution by size), but their 100,000 conformations each are finite-temperature MD frames rather than minimized geometries, and their reference energies use PBE+TS rather than QM9’s B3LYP. Energies are therefore per-molecule mean-aligned and the error is reported both in kcal/mol and as a z -score relative to each molecule’s conformational energy spread. The aligned error grows monotonically with molecular size and rigidity — from benzene (3.8 kcal/mol) through malonaldehyde and toluene to uracil (9.9 kcal/mol), i.e. with precisely the ring/aromatic/heteroatom motifs that concentrate error on QM9. Within a molecule, R does not order the per-conformation error (Spearman $\rho \approx 0$), consistent with the ANI-1x finding that the signal carries information about *which molecule* is hard rather than *which conformation*.

Table 6: Per-seed results. Best-validation MAE is on the held-out $1,000 \leq 8$ -heavy validation molecules; test MAE/RMSE are on the 109,024 9-heavy extrapolation set. Wall-clock is total training time on one H200 GPU.

Seed	best valid MAE	test MAE	test RMSE	epochs	wall-clock (s)
0	0.333	0.818	1.939	704	5114
1	0.306	0.812	1.849	745	5326
2	0.341	0.866	1.939	690	4965
mean	0.327	0.832	1.909	713	5135
ensemble	0.327	0.716	1.804	713	5135

Table 7: rMD17 conformational transfer (three-model ensemble). “Spread” is the per-molecule conformational energy standard deviation; aligned MAE is after per-molecule mean alignment; mean z is the mean $|\text{error}|$ in units of the spread; ρ is the Spearman correlation of R with per-conformation $|\text{error}|$.

Molecule	N_{heavy}	spread (kcal/mol)	aligned MAE	mean z	$\langle R \rangle$	ρ
benzene	6	2.42	3.83	1.58	3.44	-0.04
malonaldehyde	5	4.18	4.49	1.07	1.38	0.07
toluene	7	5.20	6.90	1.33	4.17	0.17
uracil	8	5.14	9.90	1.93	3.75	-0.04